

# **Rapido Ride Analytics Dashboard using Power BI**



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## **1. Project Overview**

The Rapido Ride Analytics Dashboard is an interactive Business Intelligence project developed using Microsoft Power BI to analyze ride-booking data and transform it into meaningful business insights. The primary objective of this project is to help stakeholders monitor business performance, improve operational efficiency, understand customer behavior, and evaluate revenue generation through dynamic and visually appealing dashboards.

This project demonstrates the complete Business Intelligence workflow, including data cleaning, data transformation, data modeling, DAX measure creation, dashboard design, and interactive reporting. The dataset was prepared using Power Query, where missing values were handled, data inconsistencies were corrected, and fields were standardized to ensure accurate analysis.

The dashboard is designed with a single-page interactive navigation system using Bookmarks, Selection Pane, and Navigation Buttons, allowing users to seamlessly switch between four analytical dashboards without changing report pages. This approach provides a professional user experience while maintaining a clean and organized report layout.

The project consists of four major analytical modules:

- Executive Overview – Presents key performance indicators such as total revenue, total bookings, completion rate, average ride distance, and customer ratings, along with booking status distribution and cancellation analysis.
- Operations Dashboard – Focuses on ride operations by analyzing successful rides, cancellations, pickup and drop locations, and operational performance metrics.
- Customer Dashboard – Provides insights into customer behavior, payment preferences, ride outcomes, ratings, and customer cancellation patterns.
- Revenue Dashboard – Evaluates financial performance through revenue analysis, average booking value, revenue per ride, revenue by vehicle type, payment methods, and revenue-generating locations.

Several custom DAX measures were created to calculate important business metrics, including Revenue, Revenue per Ride, Average Booking Value, Completion Rate, Customer Cancellation Percentage, Average Revenue per Customer, and other operational KPIs.

By integrating interactive slicers, KPI cards, dynamic charts, and business insight panels, the dashboard enables users to explore data efficiently and make informed decisions. This project not only demonstrates technical proficiency in Power BI but also highlights the ability to convert raw business data into actionable insights through effective data visualization and storytelling.

Overall, the Rapido Ride Analytics Dashboard serves as a comprehensive Business Intelligence solution that showcases practical skills in Power BI, DAX, Power Query, dashboard development, and business analytics while addressing real-world reporting and decision-making requirements.

## **2. Problem Statement**

Ride-hailing platforms such as Rapido generate large volumes of booking data every day, including ride details, customer information, driver activities, payment methods, ride distances, cancellations, and revenue records. While this data contains valuable business insights, it is often stored in raw tabular formats, making it difficult for stakeholders to identify trends, monitor operational performance, and make informed business decisions.

Without an interactive reporting system, organizations face challenges in tracking key performance indicators, understanding customer behaviour, identifying cancellation patterns, evaluating operational efficiency, and analysing revenue performance. Manual analysis of such large datasets is time-consuming, prone to errors, and does not provide real-time decision support.

The objective of this project is to develop an interactive Power BI dashboard that converts raw ride-booking data into meaningful visual insights. The dashboard provides a centralized platform to monitor business performance through key performance indicators (KPIs), operational analytics, customer behaviour analysis, and revenue reporting. By integrating interactive visualizations, dynamic filters, bookmarks, and navigation features, the solution enables users to explore data efficiently, identify business trends, and support data-driven decision-making.

This project aims to simplify complex operational data, improve reporting efficiency, and demonstrate how Business Intelligence tools can be used to transform raw data into actionable insights for business growth and strategic planning.

### **3. Objectives**

The primary objective of this project is to develop an interactive and user-friendly Power BI dashboard that transforms raw Rapido ride-booking data into meaningful business insights to support data-driven decision-making.

#### **Specific Objectives**

- To analyze ride booking data and present key business metrics through interactive visualizations.
- To monitor overall business performance using Key Performance Indicators (KPIs) such as Total Revenue, Total Bookings, Completion Rate, Average Booking Value, and Revenue per Ride.
- To identify operational trends by analysing successful rides, cancellations, pickup locations, drop locations, and vehicle performance.
- To understand customer behaviour by evaluating customer ratings, payment preferences, ride outcomes, and cancellation reasons.
- To analyze revenue performance by identifying high revenue-generating vehicle types, payment methods, and pickup/drop locations.
- To create a single-page interactive dashboard using Power BI Bookmarks, Navigation Buttons, and the Selection Pane for seamless navigation between multiple analytical views.
- To implement custom DAX measures for calculating business metrics and KPIs.
- To utilize Power Query for data cleaning, transformation, and preparation before analysis.
- To provide clear and actionable business insights that help stakeholders identify trends, improve operational efficiency, enhance customer satisfaction, and optimize revenue generation.
- To strengthen practical knowledge of Business Intelligence concepts, data visualization, dashboard design, and analytical problem-solving using Microsoft Power BI.

## **4. Business Requirements**

The Rapido Ride Analytics Dashboard was developed to provide stakeholders with an interactive Business Intelligence solution for monitoring ride-booking operations, customer behavior, and revenue performance. The dashboard is designed to support data-driven decision-making by presenting key business metrics in a clear and visually appealing format.

The following business requirements were identified for this project:

### **Functional Requirements**

- Develop an interactive Power BI dashboard to analyze Rapido ride-booking data.
- Provide a centralized view of overall business performance through key performance indicators (KPIs).
- Enable users to navigate seamlessly between multiple analytical dashboards using Bookmarks and Navigation Buttons on a single report page.
- Allow users to filter data dynamically using slicers such as Vehicle Type, Booking Status, and Payment Method.
- Display operational metrics including successful rides, cancellations, ride completion rate, pickup and drop locations, and ride distance.
- Analyze customer behavior through customer ratings, payment preferences, ride outcomes, and cancellation reasons.
- Monitor revenue performance using metrics such as Total Revenue, Average Booking Value, Revenue per Ride, and Average Revenue per Customer.
- Present business insights through charts, KPI cards, and summary sections to assist management in identifying trends and making informed decisions.

### **Non-Functional Requirements**

- The dashboard should provide a clean, professional, and user-friendly interface.
- Visualizations should be interactive and respond to user-selected filters.
- The report should support quick navigation between different analytical sections without changing report pages.

- The dashboard should maintain consistent formatting, color themes, and layout across all analytical views.
- Data should be accurate, cleaned, and validated before visualization.
- KPIs and calculations should be implemented using optimized DAX measures to ensure reliable business reporting.

#### Expected Business Outcomes

By implementing this dashboard, stakeholders should be able to:

- Monitor overall ride-booking performance in real time.
- Identify operational bottlenecks and major cancellation reasons.
- Understand customer behavior and payment preferences.
- Evaluate revenue performance across different vehicle types and locations.
- Identify high-demand pickup and drop locations.
- Improve operational efficiency through data-driven insights.
- Support strategic business decisions using interactive reports and visual analytics.

The dashboard serves as a comprehensive Business Intelligence solution that simplifies complex ride-booking data into meaningful and actionable business insights for management and operational teams.



## 5. Dataset Description

The dataset used for this project contains **50,000 ride-booking records** representing various aspects of a ride-hailing service similar to Rapido. The data includes information related to bookings, customers, drivers, vehicle types, ride distances, payment methods, cancellations, ratings, and revenue. It was used to analyze business performance, operational efficiency, customer behavior, and financial metrics through interactive Power BI dashboards.

Before analysis, the dataset was cleaned and transformed using **Power Query** to ensure data consistency and accuracy. Missing values, blank records, and inconsistent data entries were handled appropriately to improve the quality of the analysis.

### Dataset Summary

| Attribute                      | Description                   |
|--------------------------------|-------------------------------|
| Dataset Name                   | Rapido Ride Booking Dataset   |
| Total Records                  | 50,000                        |
| File Format                    | CSV                           |
| Domain                         | Ride-Hailing / Transportation |
| Tool Used for Data Preparation | Power Query (Power BI)        |
| Visualization Tool             | Microsoft Power BI            |

### Dataset Features

The dataset contains the following key attributes:

| Column Name  | Description                                                                 |
|--------------|-----------------------------------------------------------------------------|
| Booking ID   | Unique identifier for each ride booking                                     |
| Customer ID  | Unique identifier for each customer                                         |
| Driver ID    | Unique identifier for each driver                                           |
| Vehicle Type | Category of vehicle booked (Bike, Auto, Mini, Prime Sedan, Prime SUV, etc.) |

| Column Name                       | Description                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------|
| Booking Status                    | Status of the booking (Success, Cancelled by Customer, Cancelled by Driver, Incomplete) |
| Booking Value                     | Total fare charged for the ride                                                         |
| Ride Distance                     | Distance travelled during the ride (in km)                                              |
| Pickup Location                   | Starting location of the ride                                                           |
| Drop Location                     | Destination location of the ride                                                        |
| Customer Rating                   | Rating provided by the customer                                                         |
| Driver Rating                     | Rating provided to the driver                                                           |
| Payment Method                    | Mode of payment (Cash, UPI, Card, Wallet, etc.)                                         |
| Reason for Cancelling by Customer | Reason provided by the customer for cancelling the ride                                 |
| Reason for Cancelling by Driver   | Reason provided by the driver for cancelling the ride                                   |

### Data Preparation

The following preprocessing steps were performed before creating the dashboard:

- Removed duplicate records.
- Verified and corrected data types.
- Handled blank and null values.
- Standardized booking status values.
- Cleaned payment method fields.
- Created calculated measures using DAX.
- Validated numerical columns for accurate KPI calculations.

## 6. Data Cleaning

Data cleaning is a crucial step in the Business Intelligence process, as the quality of the dashboard depends on the accuracy and consistency of the underlying data. Before creating visualizations, the Rapido ride-booking dataset was cleaned and transformed using **Power Query** in Microsoft Power BI to ensure reliable analysis and accurate business insights.

The following data cleaning activities were performed:

### Data Cleaning Process

#### 1. Handling Missing Values

- Identified blank and null values in multiple columns.
- Replaced missing values where appropriate.
- Retained blank values only when they represented valid business scenarios (e.g., successful rides do not have cancellation reasons).

#### 2. Payment Method Standardization

- Blank payment method values associated with cancelled and incomplete bookings were replaced with "**No Payment**" to maintain consistency and improve reporting.
- This ensured accurate analysis of payment method distribution.

#### 3. Data Type Validation

Verified and corrected the data types of all columns:

- Booking ID → Text
- Customer ID → Text
- Driver ID → Text
- Booking Value → Decimal Number
- Ride Distance → Decimal Number
- Customer Rating → Decimal Number
- Driver Rating → Decimal Number
- Booking Status → Text

- Payment Method → Text

#### **4. Duplicate Record Verification**

- Checked the dataset for duplicate records.
- Confirmed that each booking was uniquely identified using the **Booking ID** field.

#### **5. Standardization of Categorical Data**

Validated and standardized categorical fields to ensure consistent reporting, including:

- Booking Status
- Vehicle Type
- Payment Method
- Pickup Location
- Drop Location
- Customer Cancellation Reasons
- Driver Cancellation Reasons

#### **6. Data Quality Checks**

Performed validation to ensure:

- No invalid booking values.
- No negative ride distances.
- Ratings were within the expected range.
- Revenue calculations were accurate.
- Booking statuses were correctly categorized.

#### **7. DAX Measure Preparation**

After cleaning the dataset, custom DAX measures were created to calculate key business metrics such as:

- Total Revenue
- Total Bookings

- Successful Rides
- Completion Rate
- Revenue per Ride
- Average Booking Value
- Average Revenue per Customer
- Customer Cancellation Percentage

### **Outcome of Data Cleaning**

The cleaned dataset provided a consistent and reliable foundation for dashboard development. By eliminating inconsistencies, handling missing values appropriately, and standardizing the data, the dashboard was able to deliver accurate visualizations and meaningful business insights for operational, customer, and revenue analysis.

## 7. Data Modeling

Data modeling is an essential phase in Power BI that establishes the relationships between data elements and enables accurate calculations and efficient report generation. In this project, the dataset was structured to support interactive dashboards and business intelligence reporting.

Since the project uses a cleaned ride-booking dataset, the data model was designed to ensure efficient analysis through calculated measures, organized fields, and optimized report performance.

### Data Modeling Process

#### 1. Data Import

The cleaned Rapido ride-booking dataset was imported into Microsoft Power BI from a CSV file using the **Get Data** feature.

#### 2. Data Preparation

After importing the dataset:

- Verified column names and data types.
- Organized numerical, categorical, and text fields.
- Ensured consistency across all attributes before visualization.

#### 3. Table Structure

The project uses a **single fact table** containing all ride-booking information. This simplified data model was sufficient because all required attributes were available within one dataset.

The fact table includes:

- Booking Details
- Customer Information
- Driver Information
- Vehicle Type
- Booking Status
- Ride Distance

- Booking Value
- Payment Method
- Customer Rating
- Driver Rating
- Pickup Location
- Drop Location
- Cancellation Reasons

#### **4. DAX Measures**

To support business analysis, several custom DAX measures were created instead of storing calculated values in the dataset. This approach improves flexibility, reduces redundancy, and enables dynamic calculations based on user-selected filters.

The major DAX measures include:

- Total Revenue
- Total Bookings
- Successful Rides
- Cancelled by Customer
- Cancelled by Driver
- Incomplete Rides
- Completion Rate (%)
- Average Booking Value
- Revenue per Ride
- Average Revenue per Customer
- Customer Cancellation Percentage
- Average Ride Distance

These measures automatically recalculate whenever users interact with slicers or filters.

## 5. Interactive Filtering

The dashboard uses interactive slicers to dynamically filter the report based on:

- Vehicle Type
- Booking Status
- Payment Method

All visualizations and KPI cards respond instantly to user selections, providing an interactive analytical experience.

## 6. Bookmark-Based Navigation

A key feature of this project is the implementation of **Bookmarks**, **Navigation Buttons**, and the **Selection Pane**.

Instead of creating multiple report pages, four analytical dashboards were organized into separate visual groups:

- Executive Overview
- Operations
- Customer
- Revenue

Bookmarks were configured to show and hide these groups, enabling seamless navigation within a single report page. This approach improves user experience while maintaining a clean and professional dashboard layout.

## Benefits of the Data Model

The implemented data model provides several advantages:

- Efficient data organization.
- Dynamic KPI calculations using DAX.
- Interactive filtering through slicers.
- Improved report performance.
- Easy maintenance and scalability.



- Seamless navigation using bookmarks.
- Accurate business reporting through reusable measures.

Overall, the data model serves as the foundation of the Rapido Ride Analytics Dashboard, enabling reliable calculations, responsive visualizations, and interactive business intelligence reporting.

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## 8. Dashboard Architecture

The Rapido Ride Analytics Dashboard was designed using a **single-page interactive architecture** to provide a seamless and user-friendly reporting experience. Instead of creating multiple report pages, the dashboard utilizes **Power BI Bookmarks**, **Navigation Buttons**, and the **Selection Pane** to switch between different analytical views while remaining on the same page.

This architecture improves report usability, maintains a consistent layout, and allows users to navigate efficiently between different business perspectives.

### Dashboard Structure

The dashboard is organized into four major analytical modules:

#### 1. Executive Overview Dashboard

The Executive Overview provides a high-level summary of the overall business performance. It enables management to quickly monitor important business metrics and understand the current state of operations.

#### Components:

- KPI Cards
  - Total Revenue
  - Total Bookings
  - Average Ride Distance
  - Average Customer Rating
  - Completion Rate
- Booking Status Distribution
- Payment Method Distribution
- Driver Cancellation Reasons
- Customer Cancellation Reasons
- Ride Funnel

- Business Insight

## **2. Operations Dashboard**

The Operations Dashboard focuses on monitoring day-to-day ride operations and operational efficiency.

### **Components:**

- Successful Rides
- Driver Cancellations
- Customer Cancellations
- Incomplete Rides
- Average Ride Distance
- Top Pickup Locations
- Top Drop Locations
- Driver Cancellation Reasons
- Booking Status by Vehicle Type
- Operational Insights

## **3. Customer Dashboard**

The Customer Dashboard analyzes customer behaviour, preferences, and satisfaction.

### **Components:**

- Average Customer Rating
- Total Customers
- Customer Cancellation Percentage
- Average Ride Distance
- Customer Rating Distribution
- Payment Method Preference
- Customer Cancellation Reasons

- Ride Outcome by Vehicle Type
- Customer Insights

#### 4. Revenue Dashboard

The Revenue Dashboard evaluates the financial performance of the business.

##### Components:

- Total Revenue
- Average Booking Value
- Revenue per Ride
- Total Bookings
- Average Revenue per Customer
- Revenue by Vehicle Type
- Revenue by Payment Method
- Top Pickup Locations by Revenue
- Top Drop Locations by Revenue
- Revenue Insights

##### Navigation Architecture

To provide a professional and interactive user experience, the dashboard implements a bookmark-based navigation system.

The navigation panel remains visible throughout the report and allows users to switch between dashboard sections using dedicated navigation buttons.

##### Navigation Flow:

Executive Overview



Operations Dashboard



Customer Dashboard



Revenue Dashboard

Each navigation button is connected to a corresponding **Bookmark**, which controls the visibility of grouped visuals using the **Selection Pane**.

### Interactive Features

The dashboard incorporates several interactive features to improve usability:

- **Bookmarks** for seamless dashboard switching.
- **Navigation Buttons** for quick access to each analytical section.
- **Selection Pane Groups** to organize dashboard components.
- **Dynamic Slicers** for filtering data by:
  - Booking Status
  - Vehicle Type
  - Payment Method
- **KPI Cards** for real-time performance monitoring.
- **Interactive Charts** that respond dynamically to slicer selections.

### Dashboard Design Principles

The dashboard was designed following modern Business Intelligence best practices:

- Clean and consistent layout.
- Uniform color palette across all dashboards.
- Easy-to-read KPI cards.
- Business-focused visualizations.

- Interactive navigation without page switching.
- Minimal clutter with maximum information visibility.
- Consistent alignment and spacing for a professional appearance.

### **Architecture Benefits**

The implemented dashboard architecture offers several advantages:

- Provides a centralized view of business performance.
- Enables fast navigation between analytical sections.
- Improves report usability through interactive bookmarks.
- Supports dynamic filtering for detailed analysis.
- Enhances decision-making through well-structured visualizations.
- Creates a professional dashboard experience similar to enterprise Business Intelligence solutions.

Overall, the dashboard architecture combines interactive navigation, structured visual design, and dynamic analytics to deliver a comprehensive and user-friendly Business Intelligence solution for ride-booking data analysis.

## 9. KPI Measures

Key Performance Indicators (KPIs) are quantitative metrics used to evaluate the overall performance of a business. In this project, several custom KPI measures were created using **DAX (Data Analysis Expressions)** to provide real-time insights into ride operations, customer behavior, and revenue performance.

These KPIs are displayed as interactive cards and automatically update based on the selected slicers and filters.

### Overview Dashboard KPIs

#### 1. Total Revenue

**Description:** Displays the total revenue generated from all completed ride bookings.

**Business Purpose:** Helps stakeholders monitor overall business earnings and financial performance.

#### 2. Total Bookings

**Description:** Represents the total number of ride bookings recorded in the dataset.

**Business Purpose:** Indicates the overall booking volume and customer demand.

#### 3. Average Ride Distance

**Description:** Calculates the average distance traveled per ride.

**Business Purpose:** Helps understand customer travel patterns and average trip length.

#### 4. Average Customer Rating

**Description:** Displays the average rating provided by customers.

**Business Purpose:** Measures overall customer satisfaction and service quality.

#### 5. Completion Rate

**Description:** Calculates the percentage of successfully completed rides.

**Business Purpose:** Evaluates operational efficiency and service reliability.

### Operations Dashboard KPIs

#### 6. Successful Rides

**Description:** Displays the total number of successfully completed bookings.

**Business Purpose:** Measures successful service delivery.

#### **7. Cancelled by Driver**

**Description:** Shows the total number of rides cancelled by drivers.

**Business Purpose:** Identifies operational challenges and driver-related issues.

#### **8. Cancelled by Customer**

**Description:** Displays the number of rides cancelled by customers.

**Business Purpose:** Helps analyze customer cancellation behavior.

#### **9. Incomplete Rides**

**Description:** Represents rides that were not completed due to operational issues.

**Business Purpose:** Highlights areas requiring operational improvement.

#### **10. Average Ride Distance**

**Description:** Displays the average distance covered across all rides.

**Business Purpose:** Assists in understanding ride patterns and operational planning.

### **Customer Dashboard KPIs**

#### **11. Average Customer Rating**

**Description:** Calculates the overall customer satisfaction score.

**Business Purpose:** Evaluates customer experience and service quality.

#### **12. Total Customers**

**Description:** Displays the total number of unique customers in the dataset.

**Business Purpose:** Measures customer reach and user base.

#### **13. Customer Cancellation Percentage**

**Description:** Calculates the percentage of bookings cancelled by customers.

**Business Purpose:** Helps identify customer cancellation trends and possible service issues.



#### **14. Average Ride Distance**

**Description:** Displays the average ride distance for customer bookings.

**Business Purpose:** Provides insight into customer travel behavior.

#### **15. Cancelled by Customer**

**Description:** Shows the total number of customer-initiated ride cancellations.

**Business Purpose:** Helps evaluate customer retention and cancellation frequency.

#### **Revenue Dashboard KPIs**

##### **16. Total Revenue**

**Description:** Displays the total revenue generated from all ride bookings.

**Business Purpose:** Measures the overall financial performance of the business.

##### **17. Average Booking Value**

**Description:** Calculates the average fare per booking.

**Business Purpose:** Helps analyze customer spending and pricing trends.

##### **18. Revenue per Ride**

**Description:** Calculates the average revenue generated per completed ride.

**Business Purpose:** Evaluates the earning potential of each booking.

##### **19. Total Bookings**

**Description:** Displays the total number of ride bookings.

**Business Purpose:** Helps compare booking volume with revenue generation.

##### **20. Average Revenue per Customer**

**Description:** Calculates the average revenue contributed by each customer.

**Business Purpose:** Measures customer value and supports customer profitability analysis.

#### **Benefits of KPI Measures**

The implemented KPI measures provide several business advantages:

- Monitor business performance in real time.
- Evaluate operational efficiency.
- Measure customer satisfaction.
- Analyze ride demand and booking trends.
- Track financial performance.
- Support strategic decision-making.
- Enable interactive analysis using Power BI slicers and filters.

These KPI measures form the foundation of the dashboard, allowing stakeholders to monitor business performance efficiently and make informed, data-driven decisions.

## 10. DAX Measures

Data Analysis Expressions (DAX) is a formula language used in Microsoft Power BI to create custom calculations, measures, and business metrics. In this project, DAX measures were developed to calculate Key Performance Indicators (KPIs) and generate dynamic insights that automatically respond to user interactions, slicers, and filters.

Using DAX enabled the dashboard to perform real-time calculations without modifying the original dataset, ensuring flexibility, accuracy, and efficient business reporting.

### DAX Measures Implemented

#### 1. Revenue

**Purpose:** Calculates the total revenue generated from all ride bookings.

Revenue =

SUM('Rapido Dataset'[Booking Value])

#### 2. Total Bookings

**Purpose:** Counts the total number of bookings in the dataset.

Total Bookings =

COUNT('Rapido Dataset'[Booking ID])

#### 3. Success Rides

**Purpose:** Counts the number of successfully completed rides.

Success Rides =

```
CALCULATE(
    COUNT('Rapido Dataset'[Booking ID]),
    'Rapido Dataset'[Booking Status] = "Success"
)
```

#### 4. Cancelled by Driver

**Purpose:** Counts the total rides cancelled by drivers.

Cancelled by Driver =

```
CALCULATE(  
    COUNT('Rapido Dataset'[Booking ID]),  
    'Rapido Dataset'[Booking Status] = "Cancelled by Driver"  
)
```

### 5. Cancelled by Customer

**Purpose:** Counts the total rides cancelled by customers.

Cancelled by Customer =

```
CALCULATE(  
    COUNT('Rapido Dataset'[Booking ID]),  
    'Rapido Dataset'[Booking Status] = "Cancelled by Customer"  
)
```

### 6. Incomplete Rides

**Purpose:** Counts the rides that remained incomplete.

Incomplete Rides =

```
CALCULATE(  
    COUNT('Rapido Dataset'[Booking ID]),  
    'Rapido Dataset'[Booking Status] = "Incomplete"  
)
```

### 7. Completion Rate

**Purpose:** Calculates the percentage of successfully completed rides.

Completion Rate =

```
DIVIDE(  
    [Success Rides],  
    [Total Bookings],
```

0

)

### 8. Average Ride Distance

**Purpose:** Calculates the average distance travelled per ride.

Average Ride Distance =

AVERAGE('Rapido Dataset'[Ride Distance])

### 9. Average Customer Rating

**Purpose:** Calculates the average customer rating.

Average Customer Rating =

AVERAGE('Rapido Dataset'[Customer Rating])

### 10. Average Booking Value

**Purpose:** Calculates the average booking amount.

Average Booking Value =

AVERAGE('Rapido Dataset'[Booking Value])

### 11. Revenue per Ride

**Purpose:** Calculates the average revenue generated per booking.

Revenue per Ride =

DIVIDE(

[Revenue],

[Total Bookings],

0

)

### 12. Total Customers

**Purpose:** Counts the total number of unique customers.

Total Customers =

DISTINCTCOUNT('Rapido Dataset'[Customer ID])

### 13. Average Revenue per Customer

**Purpose:** Calculates the average revenue generated from each customer.

Average Revenue per Customer =

DIVIDE(

[Revenue],

[Total Customers],

0

)

### 14. Customer Cancellation Percentage

**Purpose:** Calculates the percentage of bookings cancelled by customers.

Customer Cancellation % =

DIVIDE(

[Cancelled by Customer],

[Total Bookings],

0

)

### Importance of DAX in this Project

DAX measures played a significant role in enhancing the dashboard by:

- Calculating dynamic KPIs.
- Performing real-time business calculations.
- Supporting interactive filtering through slicers.
- Improving report flexibility and scalability.

- Enabling business-oriented metrics without modifying the source data.
- Providing accurate insights for operational, customer, and revenue analysis.

The implementation of DAX allowed the dashboard to deliver meaningful and responsive business intelligence, making it easier for stakeholders to monitor performance and make data-driven decisions.

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## **11. Executive Overview Dashboard**

The Executive Overview Dashboard serves as the primary dashboard of the Rapido Ride Analytics project. It provides a comprehensive summary of the organization's overall performance by presenting key business metrics and high-level insights in a single view. This dashboard is designed for managers and decision-makers who require a quick understanding of business performance without analyzing detailed operational data.

The dashboard combines Key Performance Indicators (KPIs), interactive charts, and business insights to monitor booking trends, customer satisfaction, payment preferences, cancellation patterns, and ride completion performance. Users can further explore the data using interactive slicers for Booking Status, Vehicle Type, and Payment Method.

### **Purpose**

The primary purpose of the Executive Overview Dashboard is to:

- Provide a high-level summary of overall business performance.
- Monitor key operational and financial metrics.
- Analyze booking trends and ride completion rates.
- Identify major customer and driver cancellation reasons.
- Understand customer payment preferences.
- Enable quick decision-making through interactive visualizations.

### **Key Performance Indicators (KPIs)**

The dashboard displays the following KPIs:

- Total Revenue – Represents the total revenue generated from ride bookings.
- Total Bookings – Displays the total number of ride bookings.
- Average Ride Distance – Indicates the average distance traveled per ride.
- Average Customer Rating – Measures the overall customer satisfaction level.
- Completion Rate – Shows the percentage of successfully completed rides.

These KPIs provide an instant overview of business performance and update dynamically based on the selected filters.



## **Visualizations Included**

The Executive Overview Dashboard consists of the following visualizations:

- **Ride Funnel**

Illustrates the booking journey from total bookings to successful ride completion, helping identify booking drop-offs and operational bottlenecks.

- **Ride Status Distribution**

Displays the distribution of bookings across different booking statuses, including successful rides, customer cancellations, driver cancellations, and incomplete rides.

- **Payment Method Distribution**

Shows the proportion of bookings made using different payment methods such as Cash, UPI, Card, and Wallet, enabling analysis of customer payment preferences.

- **Top Driver Cancellation Reasons**

Identifies the most common reasons why drivers cancel bookings, helping management improve operational efficiency.

- **Top Customer Cancellation Reasons**

Highlights the primary reasons customers cancel bookings, providing valuable insights into customer expectations and service improvement opportunities.

- **Key Business Insights**

A dedicated insight panel summarizes the most significant findings from the dashboard, allowing stakeholders to quickly understand important business trends.

## **Interactive Features**

The Executive Overview Dashboard includes several interactive features:

- Dynamic slicers for Booking Status, Vehicle Type, and Payment Method.
- Interactive filtering across all visualizations.
- Responsive KPI cards that update automatically based on selected filters.

- Navigation buttons for switching to Operations, Customer, and Revenue dashboards using Power BI Bookmarks.

### **Business Value**

This dashboard enables stakeholders to:

- Monitor overall business health at a glance.
- Evaluate booking performance and completion rates.
- Understand customer payment preferences.
- Identify major operational challenges through cancellation analysis.
- Make informed business decisions based on real-time insights.

### **Key Insights**

Some of the important insights obtained from the Executive Overview Dashboard include:

- The majority of ride bookings are completed successfully, indicating a strong operational completion rate.
- Driver cancellations occur more frequently than customer cancellations, suggesting opportunities to improve driver availability and operational efficiency.
- Cash and UPI are among the most preferred payment methods used by customers.
- Certain cancellation reasons occur more frequently, helping identify areas where service quality can be improved.
- The ride funnel provides a clear view of booking conversion from total requests to successfully completed rides.

## Dashboard Screenshot

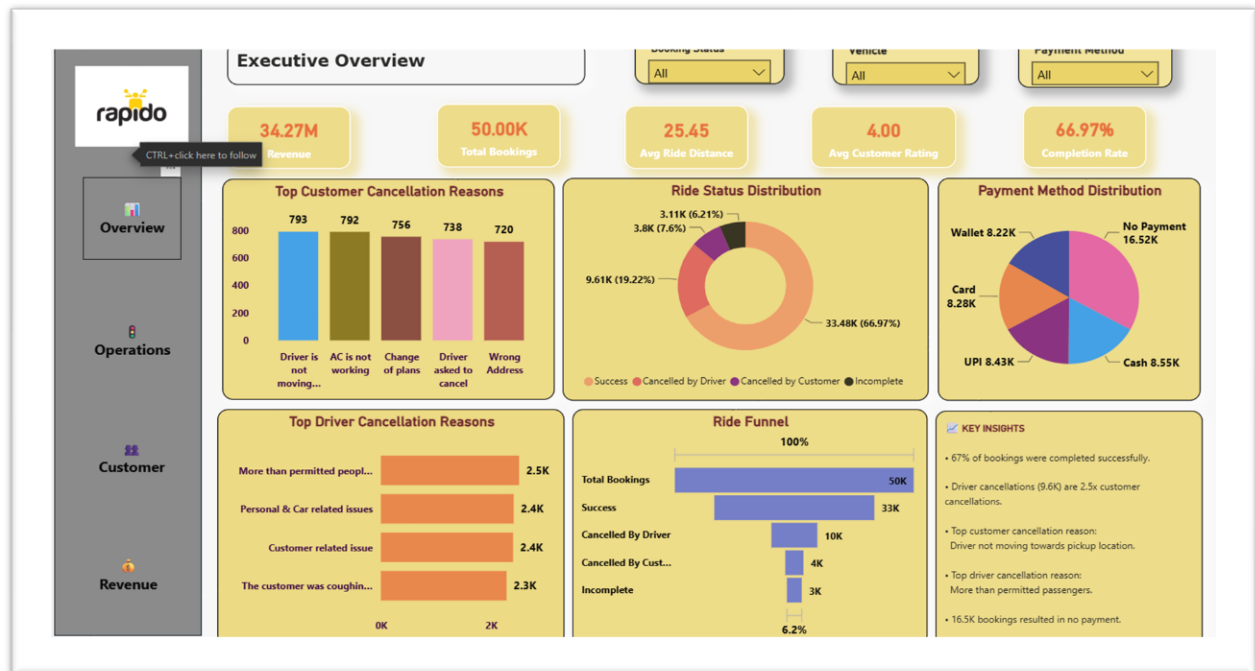


Figure 11.1: Executive Overview Dashboard

The Executive Overview Dashboard serves as the entry point to the Rapido Ride Analytics solution, providing a concise yet comprehensive summary of business performance while enabling users to navigate seamlessly to detailed analytical dashboards for Operations, Customer, and Revenue analysis.

## **12. Operations Dashboard**

The Operations Dashboard is designed to monitor the day-to-day operational performance of the ride-booking platform. It provides valuable insights into ride completion, cancellations, pickup and drop trends, and operational efficiency. This dashboard enables management and operations teams to identify bottlenecks, monitor service quality, and make informed decisions to improve the overall ride experience.

By combining Key Performance Indicators (KPIs), interactive visualizations, and business insights, the Operations Dashboard offers a comprehensive view of operational activities. Users can dynamically filter the data using slicers for Booking Status, Vehicle Type, and Payment Method, allowing detailed analysis based on specific business requirements.

### **Purpose**

The primary objectives of the Operations Dashboard are to:

- Monitor the operational performance of ride bookings.
- Track successful, cancelled, and incomplete rides.
- Identify high-demand pickup and drop locations.
- Analyze driver cancellation reasons.
- Evaluate vehicle-wise booking performance.
- Support operational decision-making through interactive analytics.

### **Key Performance Indicators (KPIs)**

The Operations Dashboard displays the following KPIs:

#### **• Successful Rides**

Represents the total number of rides that were successfully completed.

#### **• Cancelled by Driver**

Displays the total number of bookings cancelled by drivers.

#### **• Cancelled by Customer**

Represents the total number of bookings cancelled by customers.

- **Incomplete Rides**

Shows the number of rides that were not successfully completed due to operational issues.

- **Average Ride Distance**

Calculates the average distance travelled per ride, providing insights into customer travel patterns.

### **Visualizations Included**

The dashboard includes the following analytical visualizations:

- **Top Pickup Locations by Bookings**

Displays the top pickup locations with the highest booking volume. This visualization helps identify areas with high customer demand and supports resource allocation.

- **Top Drop Locations by Bookings**

Shows the most frequently selected drop locations, enabling businesses to understand travel patterns and destination hotspots.

- **Top Driver Cancellation Reasons**

Illustrates the primary reasons for driver-initiated cancellations. Understanding these reasons helps management address operational challenges and improve ride completion rates.

- **Ride Outcome by Vehicle Type**

Analyzes the distribution of booking outcomes (Successful, Cancelled by Driver, Cancelled by Customer, and Incomplete) across different vehicle categories. This visualization helps compare operational performance between vehicle types.

- **Operations Insights**

A dedicated insight panel summarizes the key findings from the dashboard, enabling stakeholders to quickly understand operational trends and performance.

### **Interactive Features**

The Operations Dashboard includes several interactive features that enhance data exploration:

- Dynamic slicers for Booking Status, Vehicle Type, and Payment Method.

- Interactive filtering across all charts and KPI cards.
- Responsive visualizations that automatically update based on user selections.
- Navigation buttons to switch seamlessly between Executive Overview, Customer, and Revenue dashboards using Power BI Bookmarks.

### **Business Value**

The Operations Dashboard provides several business benefits, including:

- Monitoring daily operational performance.
- Identifying high-demand service locations.
- Tracking cancellation trends to improve operational efficiency.
- Evaluating ride completion across different vehicle categories.
- Supporting strategic planning for driver allocation and service optimization.
- Improving customer experience by identifying operational bottlenecks.

### **Key Insights**

The Operations Dashboard provides valuable operational insights, including:

- A significant percentage of rides are completed successfully, indicating stable operational performance.
- Driver cancellations occur more frequently than customer cancellations, highlighting areas for operational improvement.
- Certain pickup and drop locations consistently record higher booking volumes, indicating demand hotspots.
- Driver cancellation reasons help identify recurring operational challenges that can be addressed to improve service quality.
- Vehicle-wise ride outcome analysis provides insights into the operational performance of different vehicle categories.

## Dashboard Screenshot

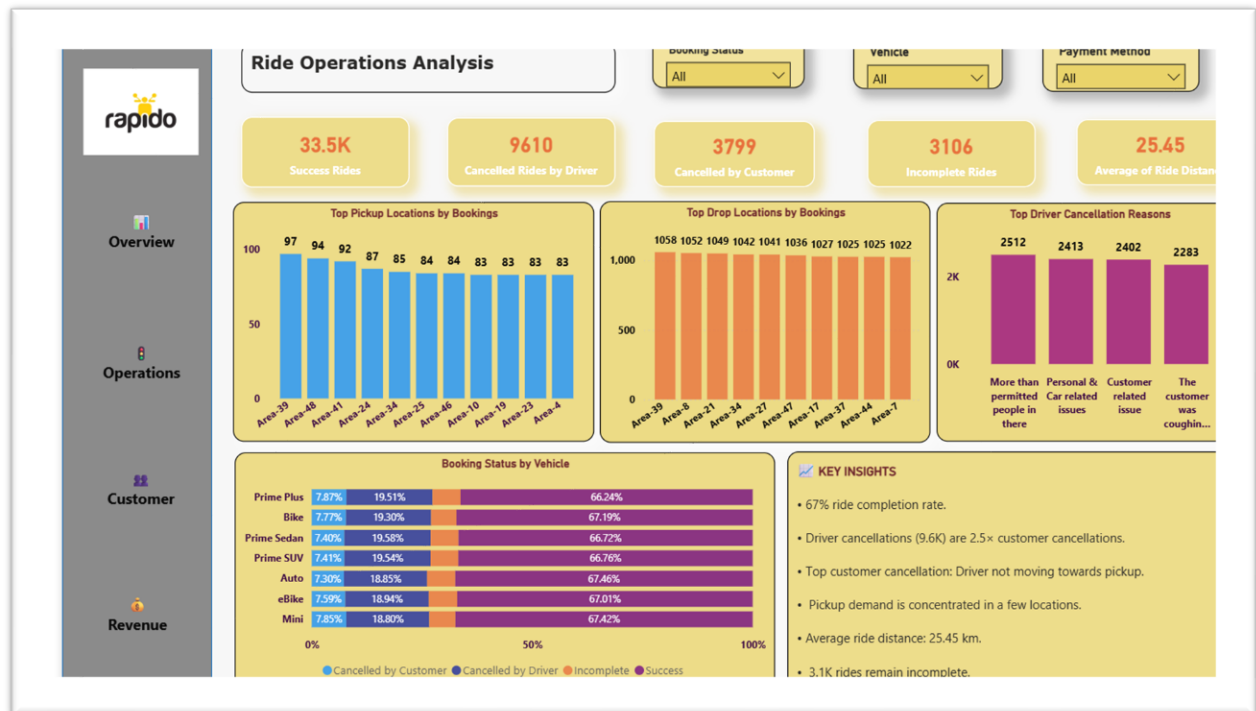


Figure 12.1: Operations Dashboard

The Operations Dashboard plays a vital role in monitoring ride-booking activities by providing real-time operational insights through interactive visualizations. It enables stakeholders to evaluate operational efficiency, identify service improvement opportunities, and make informed decisions that enhance the overall performance of the ride-booking platform.

### 13. Customer Dashboard

The Customer Dashboard focuses on analyzing customer behaviour, satisfaction, and booking preferences. It provides valuable insights into customer interactions with the ride-booking platform by examining customer ratings, payment preferences, cancellation patterns, and vehicle preferences. The dashboard helps management understand customer expectations, identify factors affecting customer satisfaction, and improve the overall customer experience.

The dashboard combines interactive KPI cards, visualizations, and business insights to present customer-related information in a clear and meaningful way. Users can filter the dashboard dynamically using **Booking Status**, **Vehicle Type**, and **Payment Method** slicers for detailed analysis.

#### Purpose

The primary objectives of the Customer Dashboard are to:

- Analyze customer satisfaction using customer ratings.
- Understand customer payment preferences.
- Identify major customer cancellation reasons.
- Evaluate ride outcomes across different vehicle types.
- Measure customer cancellation trends.
- Support customer-centric business decisions through interactive reporting.

#### Key Performance Indicators (KPIs)

The Customer Dashboard displays the following KPIs:

##### • Average Customer Rating

Displays the average rating provided by customers after completing their rides, indicating the overall quality of service.

##### • Total Customers

Represents the total number of unique customers in the dataset.

##### • Cancelled by Customer

Shows the total number of rides cancelled by customers.



- **Average Ride Distance**

Calculates the average distance travelled per ride, helping understand customer travel behaviour.

- **Customer Cancellation Percentage**

Displays the percentage of bookings cancelled by customers relative to the total number of bookings.

### **Visualizations Included**

The dashboard includes the following visualizations:

- **Customer Rating Distribution**

Displays the distribution of customer ratings, helping identify overall satisfaction levels and service quality trends.

- **Customer Payment Preference**

Illustrates the distribution of different payment methods such as Cash, UPI, Card, and Wallet, enabling analysis of customer payment behaviour.

- **Top Customer Cancellation Reasons**

Shows the most common reasons why customers cancel bookings. This visualization helps identify customer pain points and opportunities for service improvement.

- **Ride Outcome by Vehicle Type**

Analyzes booking outcomes across different vehicle categories, allowing comparison of successful rides and cancellations for each vehicle type.

- **Customer Insights**

A dedicated insights section summarizes important findings related to customer satisfaction, cancellation behaviour, payment preferences, and booking patterns.

### **Interactive Features**

The Customer Dashboard includes several interactive features to enhance user experience:

- Dynamic slicers for Booking Status, Vehicle Type, and Payment Method.

- Interactive filtering across all visualizations.
- Responsive KPI cards that automatically update based on user selections.
- Bookmark-based navigation for seamless switching between Executive Overview, Operations, Customer, and Revenue dashboards.

### **Business Value**

The Customer Dashboard provides several business benefits, including:

- Monitoring customer satisfaction through ratings.
- Understanding customer payment preferences.
- Identifying common reasons for customer cancellations.
- Evaluating customer booking behaviour across different vehicle types.
- Supporting initiatives to improve customer retention and service quality.
- Enabling data-driven customer experience improvements.

### **Key Insights**

The Customer Dashboard provides valuable insights, including:

- The average customer rating indicates a generally positive ride experience.
- Cash and UPI are among the most preferred payment methods.
- "Driver not moving towards pickup location" is one of the leading reasons for customer cancellations.
- Premium vehicle categories receive a significant share of customer bookings.
- Customer cancellation rates remain lower than successful ride completion, indicating overall customer confidence in the service.

## Dashboard Screenshot

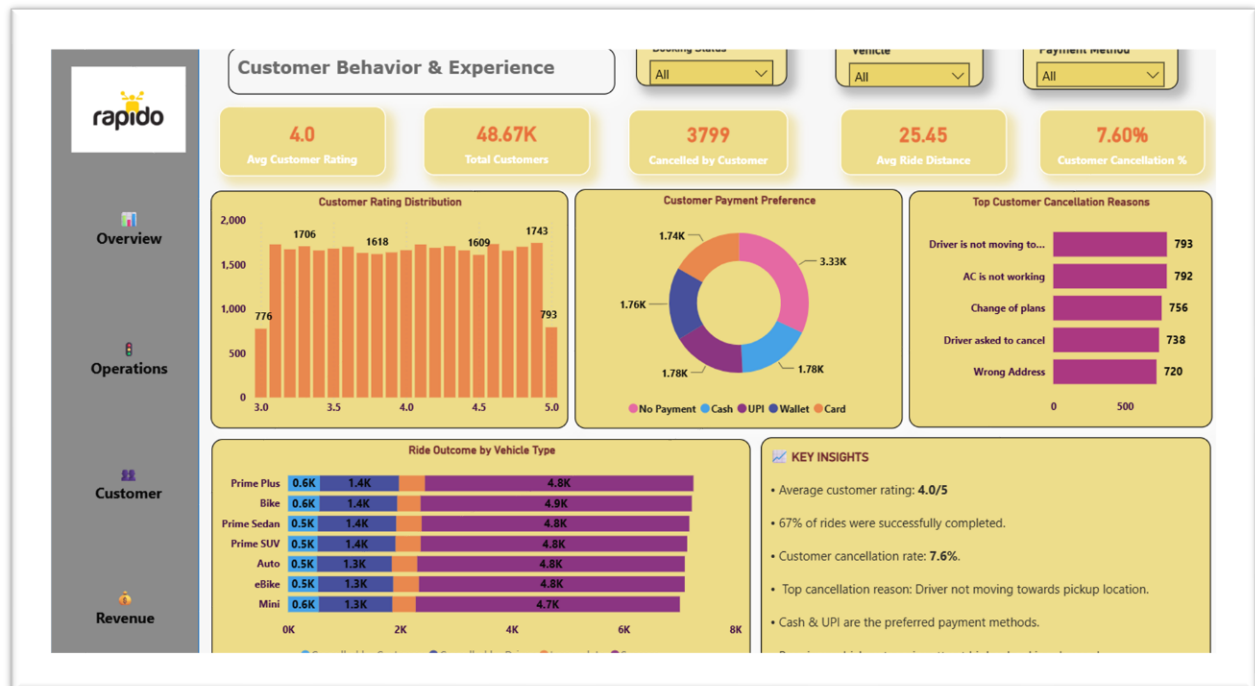


Figure 13.1: Customer Dashboard

The Customer Dashboard provides a comprehensive view of customer behaviour and satisfaction by combining operational metrics with customer-focused analytics. It enables stakeholders to understand customer expectations, identify improvement opportunities, and enhance the overall ride experience through data-driven decision-making.

## 14. Revenue Dashboard

The **Revenue Dashboard** focuses on analyzing the financial performance of the ride-booking platform. It provides a comprehensive view of revenue generation by monitoring key financial metrics, identifying high-performing locations, evaluating vehicle-wise revenue contribution, and analyzing customer payment preferences. The dashboard enables stakeholders to understand revenue trends and make informed business decisions to improve profitability and business growth.

The dashboard combines interactive KPI cards, analytical visualizations, and business insights to present financial information in a clear and meaningful format. Users can interact with the dashboard using **Booking Status**, **Vehicle Type**, and **Payment Method** slicers to perform detailed revenue analysis.

### Purpose

The primary objectives of the Revenue Dashboard are to:

- Monitor the overall financial performance of the business.
- Analyze revenue generated from ride bookings.
- Evaluate average booking value and revenue per ride.
- Measure average revenue generated per customer.
- Identify high revenue-generating pickup and drop locations.
- Compare revenue contribution across different vehicle categories.
- Analyze revenue distribution based on payment methods.
- Support strategic financial planning and business decision-making.

### Key Performance Indicators (KPIs)

The Revenue Dashboard displays the following KPIs:

#### • Total Revenue

Represents the total revenue generated from all completed ride bookings.

#### • Average Booking Value

Displays the average fare earned per booking, helping evaluate customer spending patterns.

- **Revenue per Ride**

Calculates the average revenue generated from each ride booking.

- **Total Bookings**

Displays the total number of bookings processed within the dataset.

- **Average Revenue per Customer**

Calculates the average revenue contributed by each unique customer, providing insight into customer profitability.

### **Visualizations Included**

The Revenue Dashboard includes the following visualizations:

- **Revenue by Vehicle Type**

Displays the revenue generated by each vehicle category, enabling comparison of vehicle performance and contribution to overall business revenue.

- **Revenue by Payment Method**

Illustrates the revenue distribution across different payment methods such as Cash, UPI, Card, and Wallet, helping identify customer payment preferences and payment channel performance.

- **Top Pickup Locations by Revenue**

Identifies the pickup locations generating the highest revenue, allowing businesses to recognize high-demand areas and optimize operational planning.

- **Top Drop Locations by Revenue**

Shows the destinations contributing the highest revenue, providing insights into popular travel routes and business opportunities.

- **Revenue Insights**

A dedicated insights section summarizes the most significant financial observations, allowing decision-makers to quickly understand revenue performance and business trends.

### **Interactive Features**

The Revenue Dashboard includes several interactive capabilities:

- Dynamic slicers for Booking Status, Vehicle Type, and Payment Method.
- Interactive filtering across all charts and KPI cards.
- Responsive visualizations that update automatically based on user selections.
- Bookmark-based navigation for seamless switching between Executive Overview, Operations, Customer, and Revenue dashboards.

## **Business Value**

The Revenue Dashboard provides valuable financial insights by enabling stakeholders to:

- Monitor business revenue in real time.
- Identify the most profitable vehicle categories.
- Analyze customer payment behavior.
- Recognize high revenue-generating pickup and drop locations.
- Evaluate customer spending patterns.
- Support revenue optimization and strategic business planning.

## **Key Insights**

The Revenue Dashboard provides several important business insights, including:

- The platform generates strong overall revenue through successful ride bookings.
- Revenue is distributed across multiple vehicle categories, indicating balanced demand among different ride options.
- Cash, UPI, Card, and Wallet contribute significantly to the overall revenue generated by the platform.
- Certain pickup and drop locations consistently generate higher revenue, highlighting important business hotspots.
- Average booking value and revenue per ride provide useful indicators of customer spending behavior and business profitability.
- Average revenue per customer helps evaluate customer value and supports customer-centric business strategies.

## Dashboard Screenshot

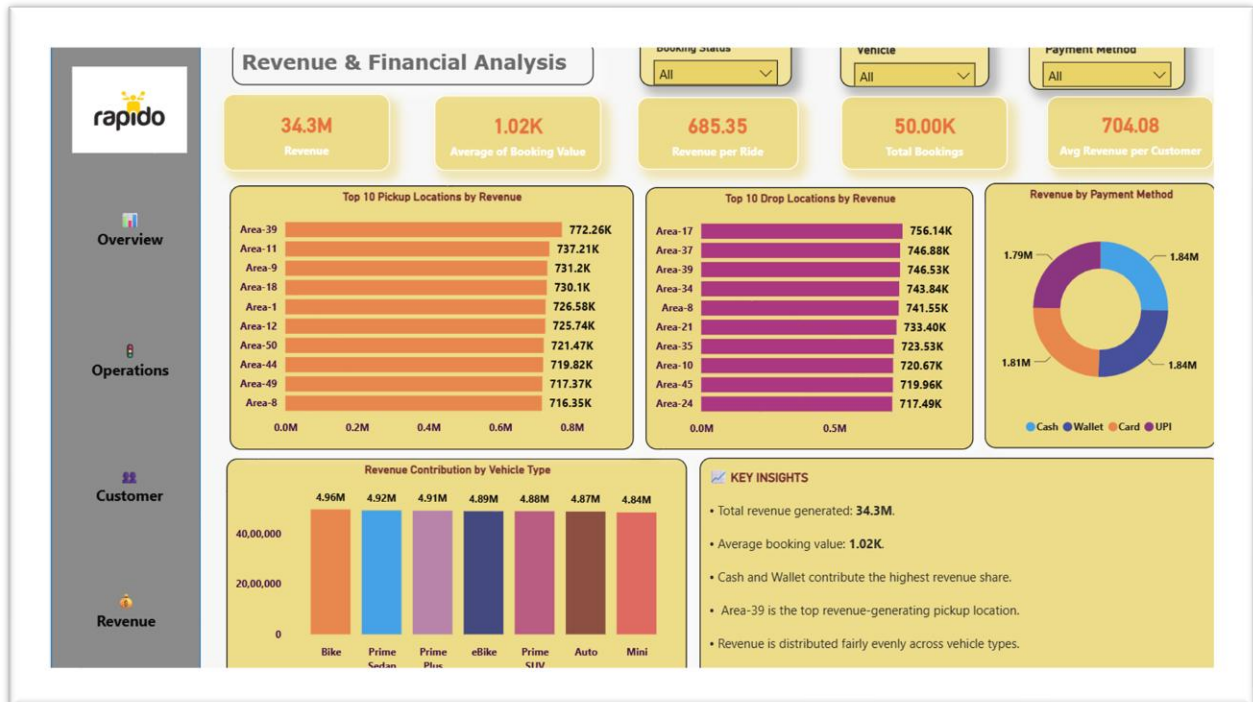


Figure 14.1: Revenue Dashboard

The Revenue Dashboard serves as the financial analysis component of the Rapido Ride Analytics solution. By integrating revenue KPIs, interactive visualizations, and dynamic filtering, it enables stakeholders to evaluate business profitability, identify revenue-driving factors, and make informed strategic decisions. Together with the Executive Overview, Operations, and Customer dashboards, it provides a comprehensive Business Intelligence solution for analyzing ride-booking performance.

## 15. Interactive Features

One of the key highlights of the **Rapido Ride Analytics Dashboard** is its interactive and user-friendly design. The dashboard was developed to provide a seamless analytical experience by allowing users to explore data dynamically without navigating through multiple report pages. Various Power BI interactive features were implemented to improve usability, flexibility, and decision-making.

### 1. Bookmark Navigation

A bookmark-based navigation system was implemented to create a **single-page interactive dashboard**.

Instead of using multiple report pages, four analytical dashboards were organized into separate groups:

- Executive Overview
- Operations Dashboard
- Customer Dashboard
- Revenue Dashboard

Each navigation button is linked to a corresponding bookmark, enabling users to switch instantly between dashboards while remaining on the same report page.

#### Benefit:

- Provides a smooth and professional user experience.
- Reduces report complexity.
- Improves dashboard accessibility.

### 2. Navigation Buttons

Interactive navigation buttons were created using Power BI's button and action features.

The navigation panel allows users to quickly access:

- Executive Overview
- Operations Dashboard



- Customer Dashboard
- Revenue Dashboard

Each button is connected to its respective bookmark, ensuring quick and seamless dashboard transitions.

**Benefit:**

- Easy report navigation.
- Application-like user interface.
- Improved report usability.

### **3. Selection Pane Groups**

The **Selection Pane** was used to organize dashboard components into logical groups.

Separate groups were created for:

- Overview Group
- Operations Group
- Customer Group
- Revenue Group
- Navigation Group

These groups are controlled through bookmarks to display only the selected dashboard while hiding the others.

**Benefit:**

- Simplifies dashboard management.
- Maintains a clean report layout.
- Enables efficient bookmark implementation.

### **4. Dynamic Slicers**

Interactive slicers were added to allow users to filter the dashboard based on different business requirements.

The dashboard includes slicers for:

- Booking Status
- Vehicle Type
- Payment Method

All KPI cards and visualizations automatically update according to the selected filters.

**Benefit:**

- Enables customized analysis.
- Provides dynamic reporting.
- Enhances user interaction.

## **5. Interactive KPI Cards**

Dynamic KPI cards were created using DAX measures to display important business metrics such as:

- Total Revenue
- Total Bookings
- Average Ride Distance
- Completion Rate
- Average Booking Value
- Revenue per Ride
- Average Revenue per Customer

These KPIs update instantly whenever users apply filters or slicers.

**Benefit:**

- Provides real-time business insights.
- Supports quick performance monitoring.

## **6. Interactive Visualizations**

The dashboard includes various interactive charts, including:

- Bar Charts
- Column Charts
- Donut Charts
- Funnel Chart
- Stacked Bar Charts
- KPI Cards

Selecting a value in one visualization automatically filters related charts, enabling deeper data exploration.

**Benefit:**

- Improves analytical capabilities.
- Supports drill-down analysis.
- Enhances business storytelling.

## **7. Dynamic Dashboard Titles**

Each analytical dashboard includes a dedicated title that changes automatically based on the selected bookmark.

The titles include:

- Executive Overview Dashboard
- Operations Dashboard
- Customer Dashboard
- Revenue Dashboard

This provides users with clear context while navigating through the dashboard.

## **8. Business Insights Panel**

Each dashboard contains a dedicated **Key Insights** section that summarizes important observations derived from the visualizations.

These insights help stakeholders quickly understand business performance without analyzing every chart individually.

**Benefit:**

- Simplifies report interpretation.
- Supports faster business decision-making.

**Benefits of Interactive Features**

The implementation of interactive features provides several advantages:

- Creates a seamless single-page dashboard experience.
- Enables quick navigation between analytical sections.
- Supports dynamic filtering and real-time analysis.
- Improves report usability and user engagement.
- Enhances decision-making through interactive visualizations.
- Delivers a professional Business Intelligence reporting experience.

Overall, these interactive features significantly improve the functionality, usability, and visual appeal of the Rapido Ride Analytics Dashboard, making it an effective Business Intelligence solution for analyzing ride-booking operations, customer behavior, and revenue performance.

## **16. Business Insights**

The Rapido Ride Analytics Dashboard provides valuable business insights by transforming raw ride-booking data into meaningful and actionable information. The interactive dashboards enable stakeholders to monitor business performance, identify operational challenges, understand customer behaviour, and evaluate revenue generation. These insights support informed decision-making and help improve overall business efficiency.

### **Executive Overview Insights**

The Executive Overview Dashboard provides a high-level summary of the organization's overall performance.

Key observations include:

- The platform maintains a strong ride completion rate, indicating efficient service delivery.
- Driver cancellations occur more frequently than customer cancellations, highlighting opportunities to improve driver availability and operational planning.
- Cash and UPI are among the most preferred payment methods, reflecting customer payment preferences.
- The booking funnel clearly illustrates the progression from total bookings to successfully completed rides, helping identify potential drop-off points.
- Customer ratings indicate an overall positive service experience.

### **Operations Insights**

The Operations Dashboard focuses on the efficiency of ride operations.

Key findings include:

- Successful rides account for the majority of total bookings, demonstrating effective operational performance.
- Certain pickup and drop locations consistently record higher booking volumes, indicating high-demand service areas.
- Driver cancellations are higher than customer cancellations, suggesting operational challenges that require attention.

- Common driver cancellation reasons provide valuable information for improving driver management and service reliability.
- Vehicle-wise ride outcome analysis helps compare operational performance across different vehicle categories.

These insights can assist management in optimizing driver allocation, reducing cancellations, and improving service efficiency.

### **Customer Insights**

The Customer Dashboard provides a comprehensive understanding of customer behavior and satisfaction.

Key findings include:

- Customer ratings indicate a generally positive ride experience.
- Customer cancellation rates remain relatively low compared to successful ride completions.
- "Driver not moving towards pickup location" is one of the most common reasons for customer cancellations.
- Cash and UPI are the preferred payment methods among customers.
- Premium vehicle categories receive significant booking demand, indicating customer preference for these services.

These insights enable the organization to improve customer experience, reduce cancellations, and strengthen customer retention strategies.

### **Revenue Insights**

The Revenue Dashboard evaluates the financial performance of the business.

Key observations include:

- The platform generates consistent revenue through successful ride bookings.
- Revenue is distributed across multiple vehicle categories, indicating balanced demand for different ride options.

- Certain pickup and drop locations contribute significantly to overall revenue, highlighting profitable service areas.
- Average booking value and revenue per ride provide useful indicators of customer spending patterns.
- Average revenue per customer helps measure customer profitability and supports revenue optimization strategies.

These insights help management identify revenue-driving factors and support strategic financial planning.

### **Overall Business Impact**

The dashboard provides a centralized Business Intelligence solution that enables stakeholders to:

- Monitor operational performance in real time.
- Identify high-demand pickup and drop locations.
- Understand customer preferences and satisfaction.
- Track key financial metrics and revenue performance.
- Analyze cancellation patterns and operational challenges.
- Support strategic planning through interactive data visualization.
- Make faster and more informed business decisions.

By integrating operational, customer, and financial analytics into a single interactive dashboard, the Rapido Ride Analytics Dashboard demonstrates how Business Intelligence tools can transform raw business data into actionable insights that improve efficiency, enhance customer experience, and support sustainable business growth.

## 17. Challenges Faced

During the development of the **Rapido Ride Analytics Dashboard**, several technical and analytical challenges were encountered. Addressing these challenges provided valuable hands-on experience in data preparation, dashboard design, DAX calculations, and interactive report development.

### 1. Handling Missing and Blank Values

One of the initial challenges was dealing with blank and missing values in the dataset. Several columns, such as **Payment Method** and **Cancellation Reasons**, contained blank entries because they were not applicable to all booking statuses.

#### Solution:

- Cleaned the data using **Power Query**.
- Replaced blank payment methods for cancelled rides with **"No Payment"**.
- Retained blank cancellation reasons where they represented valid business scenarios (e.g., successful rides).

### 2. Creating Meaningful KPIs

Selecting the right Key Performance Indicators (KPIs) was challenging because some metrics produced duplicate or less meaningful results.

#### Solution:

- Designed business-focused KPIs such as Total Revenue, Revenue per Ride, Completion Rate, Customer Cancellation Percentage, and Average Revenue per Customer.
- Replaced duplicate KPIs with more informative business metrics to improve dashboard effectiveness.

### 3. Writing DAX Measures

Developing accurate DAX measures required a good understanding of aggregation functions, filter context, and calculated measures.

#### Solution:

- Created reusable DAX measures for revenue, bookings, cancellation rates, completion rates, and customer metrics.



- Validated each measure using visual outputs to ensure calculation accuracy.

#### **4. Designing a Professional Dashboard Layout**

Creating a clean, organized, and visually balanced dashboard while presenting multiple business metrics on a single page was a significant challenge.

##### **Solution:**

- Followed dashboard design best practices.
- Maintained consistent spacing, alignment, fonts, and color themes.
- Organized KPI cards and charts to improve readability and user experience.

#### **5. Implementing Single-Page Navigation**

One of the most challenging aspects of the project was creating multiple dashboards within a single Power BI report page.

##### **Solution:**

- Used the **Selection Pane** to group dashboard components.
- Implemented **Bookmarks** to control visual visibility.
- Connected **Navigation Buttons** to bookmarks, enabling users to switch seamlessly between Executive Overview, Operations, Customer, and Revenue dashboards without changing report pages.

This feature significantly enhanced the professionalism and usability of the dashboard.

#### **6. Selecting Appropriate Visualizations**

Choosing the most suitable chart type for different business metrics required careful consideration. Some visualizations did not effectively communicate insights during the initial design.

##### **Solution:**

- Evaluated multiple chart types.
- Selected visualizations that best represented the underlying data.
- Simplified complex charts to improve clarity and business storytelling.

## 7. Generating Meaningful Business Insights

Converting raw analytical results into concise business insights required understanding the business context rather than simply displaying data.

### Solution:

- Reviewed each dashboard to identify significant trends and patterns.
- Added a dedicated **Key Insights** section to summarize important findings and support decision-making.

## 8. Maintaining Consistency Across Dashboards

Since the project consists of four analytical dashboards, maintaining a consistent design and navigation experience was essential.

### Solution:

- Used a uniform color palette, typography, spacing, and visual layout across all dashboards.
- Standardized KPI card styles and chart formatting to ensure a cohesive user experience.

## Overall Learning

Overcoming these challenges significantly enhanced my practical knowledge of **Power BI, Power Query, DAX, dashboard design, interactive reporting, and business intelligence**. The project also improved my ability to solve real-world data analysis problems, create user-friendly dashboards, and present complex information through meaningful visualizations.

Successfully resolving these challenges strengthened both my technical skills and my understanding of how Business Intelligence solutions can support effective business decision-making.

## **18. Learning Outcomes**

The development of the **Rapido Ride Analytics Dashboard** provided valuable practical experience in Business Intelligence, data analysis, and dashboard development using Microsoft Power BI. This project enhanced both technical knowledge and analytical thinking by applying theoretical concepts to a real-world business scenario.

Throughout the project, I gained hands-on experience in data preparation, visualization, and interactive report design while learning how to transform raw ride-booking data into meaningful business insights.

### **Technical Skills Acquired**

During the development of this project, I strengthened my skills in:

- Microsoft Power BI
- Power Query for data cleaning and transformation
- Data Analysis Expressions (DAX)
- Data Modeling
- Interactive Dashboard Design
- Business Intelligence Reporting
- KPI Development
- Data Visualization Best Practices
- Business Storytelling through Dashboards
- Report Optimization and Layout Design

### **Power BI Skills Improved**

This project helped me improve my understanding of several advanced Power BI features, including:

- Creating interactive KPI Cards
- Designing professional dashboard layouts
- Implementing dynamic slicers and filters

- Creating custom DAX measures
- Using Bookmarks for dashboard navigation
- Organizing visuals using the Selection Pane
- Implementing Navigation Buttons
- Building a single-page interactive dashboard
- Formatting charts and improving report aesthetics

### **Analytical Skills Developed**

Working on this project enhanced my ability to:

- Analyze large datasets efficiently.
- Identify meaningful business trends.
- Convert raw data into actionable insights.
- Select appropriate visualizations for different business scenarios.
- Design dashboards that support data-driven decision-making.
- Interpret operational, customer, and financial performance metrics.

### **Problem-Solving Skills**

Throughout the project, I learned how to:

- Handle missing and inconsistent data.
- Resolve data quality issues using Power Query.
- Create dynamic calculations using DAX.
- Improve dashboard usability through interactive navigation.
- Optimize report design for better user experience.
- Translate business requirements into effective analytical solutions.

### **Business Intelligence Understanding**

This project provided a deeper understanding of how Business Intelligence supports organizations by:

- Monitoring key business performance indicators.
- Identifying operational inefficiencies.
- Understanding customer behavior.
- Evaluating financial performance.
- Supporting strategic planning through data-driven insights.
- Improving decision-making with interactive dashboards.

### **Key Takeaways**

Some of the most valuable lessons learned from this project include:

- Data cleaning is essential for accurate reporting and reliable analysis.
- Well-designed KPIs help stakeholders monitor business performance effectively.
- Interactive dashboards significantly improve user experience and decision-making.
- DAX measures enable dynamic and flexible business calculations.
- Bookmarks and Navigation Buttons can create a professional single-page dashboard experience.
- Effective data visualization is not only about displaying data but also about communicating meaningful business insights.

### **Overall Learning**

The Rapido Ride Analytics Dashboard significantly improved my practical knowledge of **Business Intelligence, Power BI, Power Query, DAX, and dashboard development**. More importantly, it enhanced my ability to think analytically, solve business problems using data, and present complex information in a clear and interactive manner.

This project has strengthened my confidence in developing professional Business Intelligence solutions and has prepared me for real-world Data Analyst and Business Intelligence roles.

## **19. Future Enhancements**

The current Rapido Ride Analytics Dashboard provides comprehensive insights into ride bookings, operational performance, customer behavior, and revenue analysis. However, there are several opportunities to further enhance the dashboard by incorporating advanced Business Intelligence features, predictive analytics, and real-time data integration.

The following enhancements can be implemented in future versions of the project:

### **1. Real-Time Data Integration**

Currently, the dashboard is based on a static dataset. Future versions can be connected to a live database or cloud-based data source such as SQL Server, Azure SQL Database, or Microsoft Fabric to enable real-time data updates and continuous monitoring.

#### **Benefits:**

- Live business monitoring
- Automatic data refresh
- Real-time operational insights

### **2. Predictive Analytics**

Machine Learning models can be integrated with Power BI to forecast future business trends, including:

- Ride demand prediction
- Revenue forecasting
- Customer cancellation prediction
- Driver availability forecasting

#### **Benefits:**

- Better resource planning
- Improved demand forecasting
- Data-driven business strategies

### **3. Customer Segmentation**

Future enhancements can include customer segmentation based on booking behavior, ride frequency, payment preferences, and spending patterns.

Possible customer segments include:

- Frequent Customers
- Premium Customers
- Occasional Riders
- High-Value Customers

**Benefits:**

- Personalized marketing campaigns
- Improved customer retention
- Better customer relationship management

#### **4. Driver Performance Dashboard**

A dedicated dashboard can be developed to monitor driver performance using metrics such as:

- Driver Ratings
- Ride Completion Rate
- Cancellation Rate
- Total Trips Completed
- Revenue Generated per Driver

**Benefits:**

- Performance evaluation
- Driver incentive planning
- Operational efficiency improvement

#### **5. Geographical Analysis**

Integrating Power BI Maps would enable geographical visualization of ride demand.

Possible analyses include:

- Pickup Hotspots
- Drop Hotspots
- Revenue by Region
- High-Demand Areas

**Benefits:**

- Better service planning
- Improved driver allocation
- Regional performance analysis

## **6. Mobile-Responsive Dashboard**

The dashboard can be optimized using Power BI Mobile Layout to provide a seamless experience on smartphones and tablets.

**Benefits:**

- Improved accessibility
- Better user experience
- Anywhere, anytime monitoring

## **7. Advanced Drill-Through Reports**

Additional drill-through pages can be implemented to provide detailed analysis for specific:

- Customers
- Drivers
- Vehicle Types
- Locations
- Revenue Categories

**Benefits:**

- Detailed investigation
- Improved analytical capabilities



- Enhanced decision support

## **8. Automated Alerts and Notifications**

Power BI alerts can be configured to notify stakeholders when predefined business conditions are met.

Examples include:

- High cancellation rate
- Low completion rate
- Revenue below target
- Unusual booking patterns

### **Benefits:**

- Faster response to business issues
- Proactive decision-making
- Improved operational monitoring

## **9. AI-Powered Insights**

Future versions can leverage Power BI's AI capabilities to automatically identify trends, anomalies, and hidden patterns in ride-booking data.

Examples include:

- Smart Narratives
- Key Influencers Analysis
- Decomposition Tree
- Anomaly Detection

### **Benefits:**

- Deeper business insights
- Reduced manual analysis
- Improved strategic planning

## **10. Enhanced Security and User Access**

Role-Level Security (RLS) can be implemented to restrict data access based on user roles.

Example roles:

- Administrator
- Operations Manager
- Finance Manager
- Customer Support Team

### **Benefits:**

- Improved data security
- Controlled data access
- Secure enterprise reporting

### **Future Vision**

The long-term vision of this project is to evolve the Rapido Ride Analytics Dashboard into a complete **Business Intelligence platform** capable of providing real-time monitoring, predictive analytics, customer intelligence, and automated decision support.

By integrating advanced Power BI capabilities, cloud-based data sources, and AI-driven analytics, the dashboard can become a scalable enterprise solution that supports strategic planning, operational excellence, customer satisfaction, and sustainable business growth.

## **20. Conclusion**

The Rapido Ride Analytics Dashboard successfully demonstrates how Business Intelligence tools can transform raw ride-booking data into meaningful, interactive, and actionable business insights. By leveraging Microsoft Power BI, Power Query, and DAX, the project provides a comprehensive solution for analyzing operational performance, customer behavior, and revenue generation.

The dashboard was designed using a single-page interactive architecture with Bookmarks, Navigation Buttons, and the Selection Pane, enabling users to seamlessly navigate between the Executive Overview, Operations, Customer, and Revenue dashboards without changing report pages. This approach enhances usability and delivers a professional reporting experience.

Throughout the project, data was cleaned, transformed, and analyzed to create meaningful Key Performance Indicators (KPIs) and visualizations that support data-driven decision-making. The dashboard enables stakeholders to monitor booking performance, identify operational challenges, understand customer preferences, evaluate financial performance, and recognize high-demand locations through interactive reports and dynamic filtering.

In addition to strengthening technical skills in Power BI, Power Query, DAX, and data visualization, this project enhanced my understanding of Business Intelligence, dashboard design, data storytelling, and analytical problem-solving. It provided practical experience in converting business requirements into an effective analytical solution while following industry-standard dashboard design practices.

Overall, the Rapido Ride Analytics Dashboard demonstrates my ability to design professional Business Intelligence solutions that combine technical expertise with business understanding. This project serves as a strong portfolio piece, showcasing my capability to develop interactive dashboards that support strategic decision-making and deliver valuable business insights.